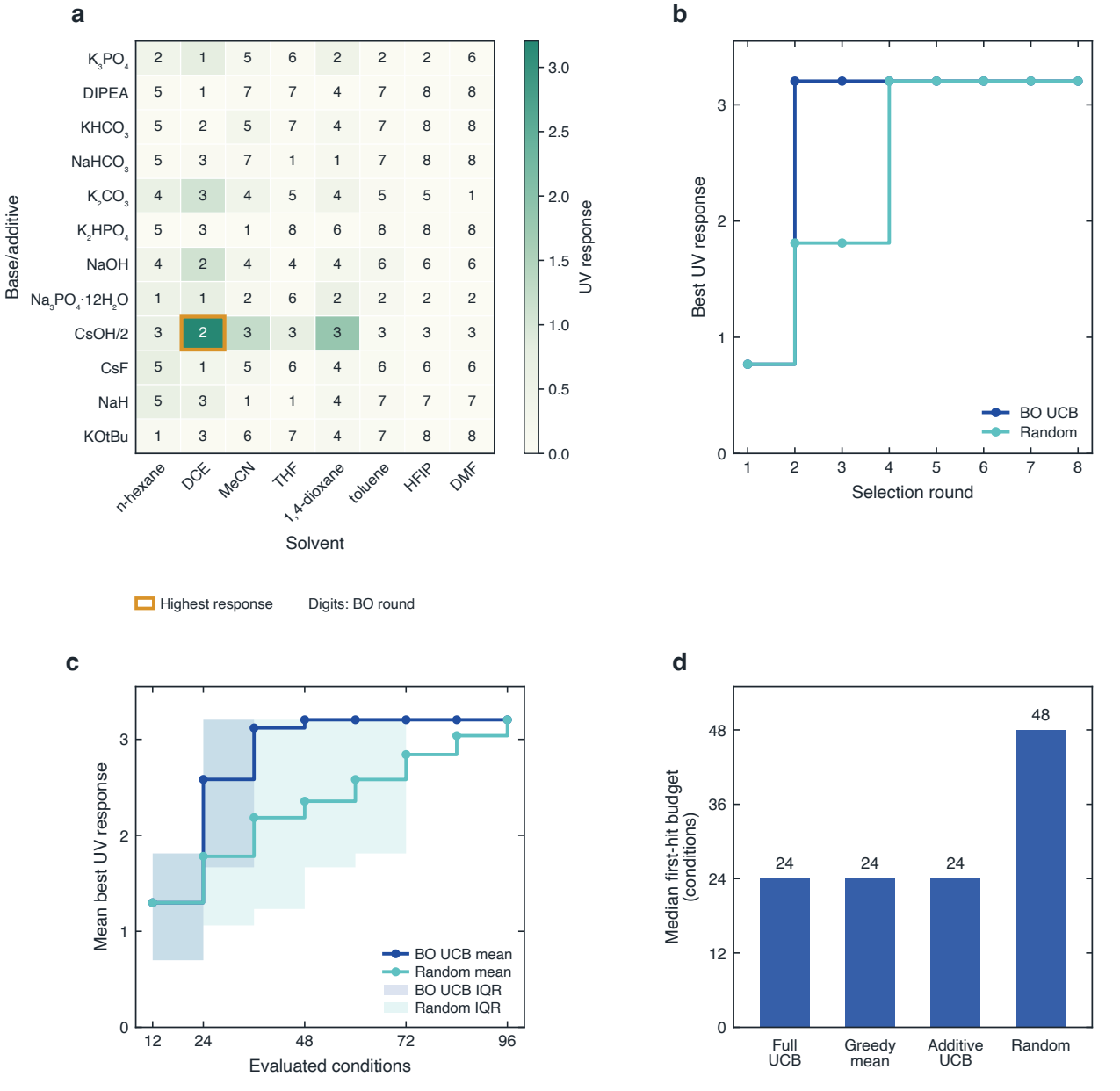
a
Base/additive
K3PO4 2 1 5 6 2 2 2 6
DIPEA 5 1 7 7 4 7 8 8
KHCO3 5 2 5 7 4 7 8 8
NaHCO3 5 3 7 1 1 7 8 8
K2CO3 4 3 4 5 4 5 5 1
K2HPO4 5 3 1 8 6 8 8 8
NaOH 4 2 4 4 4 6 6 6
Na3PO4·12H2O 1 1 2 6 2 2 2 2
CsOH/2 3 2 3 3 3 3 3 3
CsF 5 1 5 6 4 6 6 6
NaH 5 3 1 1 4 7 7 7
KOtBu 1 3 6 7 4 7 8 8
n-hexane
DCE
MeCN
THF
1,4-dioxane
toluene
HFIP
DMF
Solvent
UV response
3.0
2.5
2.0
1.5
1.0
0.5
0.0
Highest response
Digits: BO round
b
Best UV response
Selection round
BO UCB
Random
c
Mean best UV response
Evaluated conditions
BO UCB mean
Random mean
BO UCB IQR
Random IQR
d
Median first-hit budget (conditions)
24
24
24
48
Full UCB
Greedy mean
Additive UCB
Random